

# House Price Prediction – Project Summary

## Overview

The objective of this project was to develop a machine learning model capable of predicting house prices using various property-related features. The dataset included information such as area, number of bedrooms, bathrooms, parking spaces, furnishing status, air conditioning, and location preferences. The project involved data exploration, preprocessing, model development, performance evaluation, and visualization.

## Data Preparation

The dataset was loaded and inspected for missing values, duplicates, and data types. Since the dataset contained categorical attributes, these features were converted into numerical form using one-hot encoding. The cleaned dataset was then divided into training and testing sets using an 80:20 ratio.

## Model Development

Two regression algorithms were implemented:

1. Linear Regression
2. Random Forest Regressor

Both models were trained using the same dataset and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and  $R^2$  Score.

## Results

### Linear Regression

- MAE: 970,043
- RMSE: 1,324,507
- $R^2$  Score: 0.653

### Random Forest Regressor

- MAE: 1,021,546
- RMSE: 1,400,566
- $R^2$  Score: 0.612

Among the two models, Linear Regression achieved better overall performance with a higher  $R^2$  score and lower prediction errors.

## **Key Findings**

The analysis showed that property area, number of bathrooms, parking facilities, air conditioning, and preferred location significantly influence house prices. Larger homes with additional amenities generally have higher market values. The visualizations also revealed clear relationships between several housing features and the target variable.

## **Conclusion**

This project demonstrated the use of machine learning techniques for real estate price prediction. Linear Regression proved to be the most suitable model for this dataset, explaining approximately 65% of the variation in house prices. The findings can support real estate businesses in estimating property values more accurately and making informed pricing decisions based on key property characteristics.